Loading and Performance Activity

Appendix 2

Angle of bank Φ	Load factor n
0°	1.0
10°	1.015
30°	1.154
45°	1.414
60°	2.000
70°	2.923
80°	5.747
85°	11.473
90°	∞

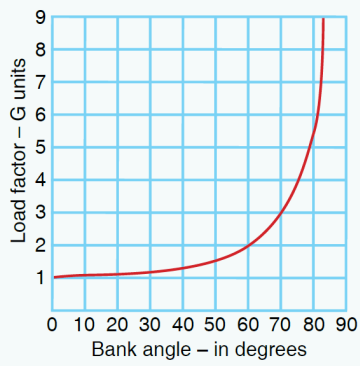


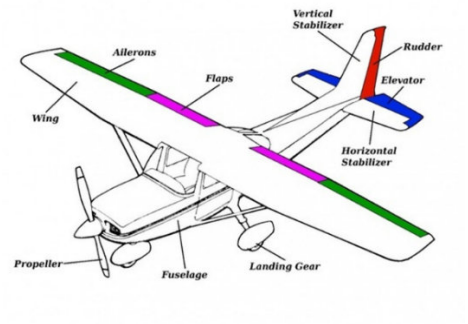
Figure 2. Load Factor Chart.

1. You normally fly your sUAS with a 1.5lbs. battery but you have just upgraded it to a 2lbs. battery. What was the load factor at 1lbs. in a 60 degree bank compared to the load factor with your new battery in a 60 degree bank?

Load factor at 1.5lbs:_____

Load factor at 2 lbs:_____

2. What happens to the load factor on an aircraft as you bank past 60 degrees?
3. Which control surface controls Yaw? Pitch? Roll?
4. What are the purpose of flaps?



5. As you increase your angle of bank in a turn, does your stall speed increase or decrease?
6. What are some methods for determining altitude of a sUAS if you don't have an altimeter installed?
7. The runway you are launching from has an uphill slope. How will this affect your take-off? How will it affect landing?
8. How much weight would a structure on your drone need to withstand if it weights 20lbs and you are in a 45 degree banked turn?
9. What is a stall? Define it in terms your grandmother would understand.
10. What is a spin? What is the procedure for recovering from a spin?